CSE7302- University Project
Review-3

FarmEkoX: Smart Crop Advisory System for Small and Marginal Farmers.

Batch Number: CSG_42

Roll Number	Student Name
20221CSG0153	NAVEEN M
20221LCG0004	AVINASH N
20221LCG0005	KURUBA ARUN KUMAR

Under the Supervision of,
Ms. Shet Reshma Prakash
Assistant Professor
School of Computer Science and Engineering
Presidency University

Name of the Program: CSG
Name of the HoD: Dr. Pallavi R
Name of the Program Project Coordinator: Ms. Benitha Christinal J

Name of the School Project Coordinators: Dr. Sampath A K

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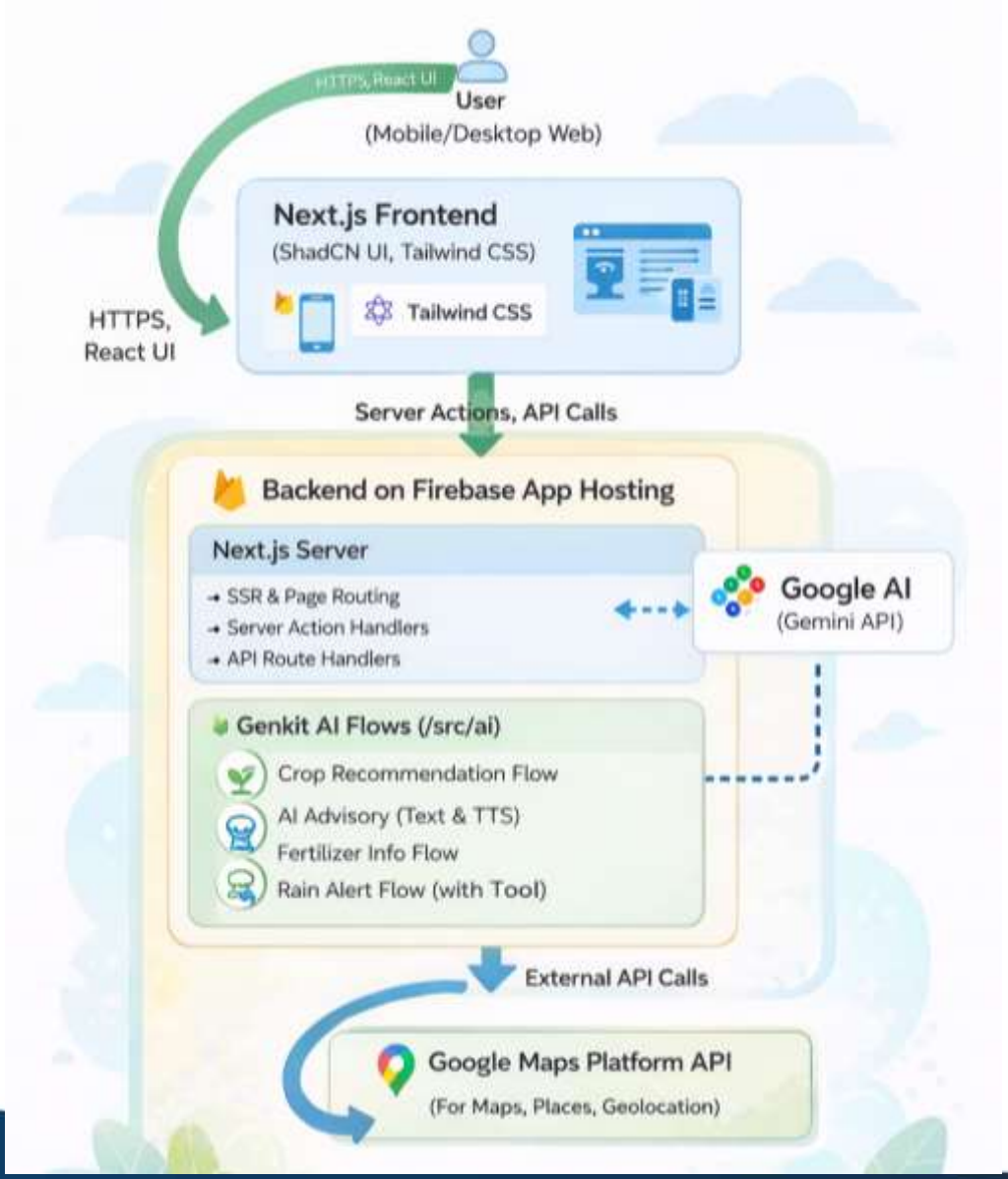
Architecture Diagram

- This diagram shows the overall architecture of the FarmEkoX system, including the user interface, backend processing, AI integration using Gemini API, and external services like Google Maps.



Algorithm Overview of FarmEkoX

FarmEkoX uses an intelligent decision-support algorithm to analyze farmer inputs and generate personalized agricultural recommendations.

The algorithm processes different types of data including:

- Farmer location information
- Weather forecast data
- Crop and fertilizer information
- Farmer advisory queries

The system integrates **AI models, rule-based logic, and external APIs** to produce accurate farming guidance.

This helps farmers improve crop productivity and reduce unnecessary resource usage.

System Working Algorithm

The system follows a structured workflow to process farmer inputs.

Process Steps

- The farmer opens the FarmEkoX web application using mobile or desktop.
- The user interacts with the dashboard interface.
- Farmer inputs data such as location, crop queries, or fertilizer image.
- The frontend sends requests to the backend server.
- The backend processes requests using Firebase services and AI modules.
- External APIs provide additional information such as maps or location data.
- AI models analyze the data and generate recommendations.
- The final advisory results are displayed to the farmer.

Technology Implementation in FarmEkoX

Frontend

- Next.js
- React.js
- Tailwind CSS

Responsible for displaying the user interface and collecting farmer inputs.

Backend

- Firebase App Hosting
- Node.js server logic

Handles request processing and application logic.

AI Integration

- Genkit Framework
- Google Gemini API

Provides intelligent crop advisory and AI-powered responses.

External API Integration

- Google Maps Platform API

Provides location services and nearby agricultural resources.

FarmEkoX Project File Structure

The FarmEkoX project uses a **modular folder structure** for better development and organization.
src/app – Next.js pages and routing *(handles application screens)*
src/components – Reusable UI components *(buttons, layouts, cards)*
src/firebase – Firebase configuration *(backend services and database)*
src/ai – AI logic and Genkit flows *(crop advisory processing)*
src/hooks – Custom React hooks *(shared application logic)*
src/lib – Utility functions and translations *(helper libraries)*

Github Link

Github link:
https://github.com/arunalbert777/Farmekox6
Or
Click here

Thank
You!